

Synergistic Effects of Rice Husk Ash and Polypropylene Fiber on the Performance of Modified Expanded Polystyrene Cement Mortar

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ABSTRACT

Lightweight cement mortar produced with Expanded Polystyrene (EPS) is an effective approach for reducing the self-weight of cement-based elements; however, EPS inclusion commonly reduces strength and toughness due to poor interfacial bonding, high buoyancy, and weak stress transfer within the cement matrix. This study investigates the synergistic effects of Rice Husk Ash (RHA) and Polypropylene (PP) fiber on the performance of modified EPS cement mortar. Surface-coated EPS beads were used as lightweight aggregate, RHA was incorporated as a pozzolanic cement replacement, and PP fibers were added as micro-reinforcement to improve crack resistance. The mix proportions were developed using Response Surface Methodology (RSM) with a Face-Centered Central Composite Design (CCF), while the Absolute Volume Method was used for accurate batching of the lightweight and pozzolanic constituents. The performance of the mortar was evaluated through compressive strength, flexural strength, density, water absorption, toughness, and microstructural observations.

The results showed that EPS replacement reduced mortar density but caused a reduction in mechanical strength due to its void-like behavior. The incorporation of RHA improved the cementitious matrix through pozzolanic reaction and pore refinement, with 10% RHA giving the most effective strength recovery. Surface coating of EPS significantly improved the interfacial transition zone by promoting mechanical interlocking between the EPS particles and cement paste. The addition of PP fibers further enhanced flexural performance and toughness by bridging microcracks and controlling brittle failure, with an optimum fiber dosage identified before the onset of fiber balling and strength reduction. Overall, the combined use of modified EPS, RHA, and PP fiber produced a lightweight cement mortar with improved mechanical performance, reduced water absorption tendency, and enhanced toughness. This study confirms that a properly optimized ternary mortar system can balance density reduction, strength development, and crack resistance for lightweight cement-based construction applications.

Keywords— Modified EPS cement mortar, Rice Husk Ash, Polypropylene fiber, lightweight mortar, Response Surface Methodology, compressive strength, flexural strength, density, water absorption, toughness.

I. INTRODUCTION

The construction industry is increasingly shifting from traditional in-situ wet construction to factory-controlled off-site prefabrication because it improves quality, reduces material waste, and shortens construction time [1]. In high-density residential and commercial projects, prefabricated non-load-bearing wall panels are widely used as internal partitions and façade cladding. These panels are attractive because they provide good dimensional accuracy, smooth finished surfaces, and faster installation, which is important in urban areas facing labor shortages and rapid housing demand [2, 3].

However, conventional cement mortar and fine concrete have high density, generally around 2200–2400 kg/m³, which increases the dead weight of precast wall elements. Heavy panels require high-capacity transport and lifting equipment, increase handling stresses during demolding, stacking, and installation, and may cause early cracking or mechanical failure before the panel reaches full strength [4]. Therefore, developing lightweight mortar with a proper balance between strength, density, and toughness is essential for prefabricated wall applications [5].

Expanded Polystyrene (EPS) beads can reduce mortar density because of their very low weight and high air content, but raw EPS has a smooth, hydrophobic surface that weakens bonding with cement paste [2, 3]. Surface coating with polymer binder and fine M-sand improves mechanical interlocking and reduces segregation [3]. Rice Husk Ash (RHA) is used as a pozzolanic material that reacts with calcium hydroxide and forms additional C-S-H gel, improving the cement matrix around EPS particles [6]. Polypropylene (PP) fibers are added to bridge cracks, control shrinkage, and improve post-cracking toughness [7]. Thus, the combined use of coated EPS, RHA, and PP fiber can produce a lightweight, stronger, and tougher cement mortar suitable for precast partition panels [8]. This ternary approach addresses density reduction, matrix refinement, and crack resistance in one optimized mortar system. While EPS lightweight mortars, RHA-modified pastes, and PP-fiber-reinforced composites have each been studied individually, their combined optimization as a single ternary system — balancing density, strength, and toughness together through Response Surface Methodology — has not been reported, and is the specific contribution addressed here.

II. MATERIALS AND EXPERIMENTS PROCEDURE

A. Materials and Apparatus

The main materials used in this study were Ordinary Portland Cement, manufactured sand, coated Expanded Polystyrene beads, Rice Husk Ash, polypropylene fibers, and superplasticizer. Tokyo Super Ordinary Portland Cement of strength class 42.5 R, conforming to SLS 107:2015, was used as the binder. The cement had a specific gravity of 2.97, Blaine fineness of $3500 \text{ cm}^2/\text{g}$, standard consistency of 30%, initial setting time of 160 min, and final setting time of 600 min. Its 2-day and 28-day compressive strengths were 25 N/mm^2 and 55 N/mm^2 , respectively [9].



Figure 1 : M Sand

Manufactured sand obtained from crushed metamorphic rock was used as the fine aggregate. Its grading, specific gravity, water absorption, bulk density, and moisture content were evaluated using standard ASTM procedures. Expanded Polystyrene beads were used as lightweight aggregate to reduce mortar density. Since raw EPS has poor bonding with cement paste, the beads were coated using Polyvinyl Acetate (PVAc) binder and fine M-sand to improve surface roughness and mechanical interlocking [10].



Figure 2 : Coated EPS

Rice Husk Ash was used as a supplementary cementitious material. It was sieved through $600 \mu\text{m}$, ball milled at about 96 rpm, and its final fineness was checked using a $53 \mu\text{m}$ wet sieve. Polypropylene fibers of 6 mm length, $25 \mu\text{m}$ diameter, 0.91 g/cm^3 density, and 560 MPa tensile strength were added to improve crack resistance and toughness. FAIRFLO S superplasticizer, a modified sulphonated naphthalene polymer, was used at 0.6–1.5% by cement weight to maintain mortar workability at $110 \pm 5 \text{ mm}$ flow.

The main apparatus included a laboratory ball mill, mechanical pan mixer, standard $40 \times 40 \times 160 \text{ mm}$ steel prism moulds, vibrating table, pycnometers, graduated cylinders, sieve shaker, drying oven, muffle furnace, mortar flow table, Universal Testing Machine, and SEM–EDS equipment for microstructural analysis

B. Experimental Procedure

a. Material Characterization and Processing

All raw materials were characterized before mix preparation to confirm their suitability. OPC specific gravity was determined using kerosene pycnometer. M-sand was tested for particle size distribution according to ASTM C136[11], and its grading was checked with BS 882[12] Zone M requirements. Specific gravity, water absorption, bulk density, and moisture content were also measured using standard ASTM methods. RHA was evaluated for chemical composition using SEM–EDS, loss on ignition at 750°C for 2 hours, and fineness using a $53 \mu\text{m}$ wet sieve. EPS beads



Figure 4 : Processed RHA

were tested using a hydrostatic water displacement method due to their very low density.

RHA was dry-sieved through $600 \mu\text{m}$ mesh and ball milled at about 96 rpm for 2 hours. For EPS coating, 100 g of raw EPS was mixed with 250 g of PVAc emulsion for 5 minutes, followed by fine M-sand addition and another 5 minutes of mixing. The coated EPS beads were dried for 24 hours before use.

b. Mix Design and Mortar Preparation

Mix proportions were developed using Response Surface Methodology (RSM) with a Central Composite Design (CCD). The Absolute Volume Method was used to ensure that all constituents, water, and air produced 1.0 m^3 of mortar. In Phase 1, the water-to-binder and sand-to-binder ratios were varied to identify the optimum plain mortar matrix. In later phases, the baseline values were fixed at $w/b = 0.425$ and $s/b = 2.25$, with 11 mix variations in each CCD phase.

Mortar was mixed in a mechanical pan mixer. OPC and RHA were mixed dry for 2 minutes, followed by M-sand for another 2 minutes. About 80% of the water with superplasticizer was then added and mixed for 1.5 minutes. Coated EPS was added with the remaining 20% water and mixed for 3 minutes. For fiber mixes, 6 mm PP fibers were added gradually over 2 minutes, followed by a final 2-minute mixing. The flow was maintained at $110 \pm 5 \text{ mm}$.

c. Casting, Curing, and Testing

Fresh mortar was cast into $40 \times 40 \times 160 \text{ mm}$ steel prism moulds in layers and compacted on a vibrating table at 50 Hz for 30 seconds per layer. Specimens were covered for 12–24 hours, demolded, and water-cured until 28 days.



Figure 3 : Casted Prisms

At 28 days, flexural strength was tested using a three-point bending setup with a 100 mm span and $50 \pm 10 \text{ N/s}$ loading rate. The broken halves were then tested for compressive strength using a $40 \times 40 \text{ mm}$ loading area at $2400 \pm 200 \text{ N/s}$. Water absorption and apparent porosity were determined by oven drying at $105 \pm 5^\circ\text{C}$ for 24 hours, followed by water immersion and mass measurements.

d. Microstructural Analysis

Selected fractured samples containing the aggregate–paste interface were prepared for SEM–EDS analysis. Hydration was arrested using isopropyl alcohol, and the samples were dried, epoxy mounted, polished, carbon taped, and gold coated. SEM images and EDS line scans were used to examine the interfacial transition zone and calcium-to-silicon ratio near coated EPS particles.

III. CALCULATIONS

The mix proportions were calculated using the absolute volume method, where the total volume of mortar was taken as the sum of the absolute volumes of binder, water, fine aggregate, lightweight aggregate, fibers, and entrapped air [13]. In this study, a 1% entrapped air allowance was

considered; therefore, the total volume of solid materials and mixing water was maintained at 0.99 m³ for a theoretical 1 m³ batch. The calculated quantities were then scaled down to the required laboratory batch volume of 0.00096 m³ for casting 40 × 40 × 160 mm mortar prisms.

Before mix design, the material properties required for volume conversion were calculated using standard procedures. The specific gravity values of cement, RHA, M-sand, EPS, and superplasticizer were used to convert material mass into absolute volume [14]. The particle size distribution of M-sand was obtained from sieve analysis and checked against the BS 882 grading limits [15], [16]. The water absorption, bulk density, and field moisture content of M-sand were calculated using standard aggregate testing procedures [15]-[17]. Since the measured field moisture content of sand was higher than its water absorption capacity, the excess free water in sand was deducted from the mixing water to maintain the selected water-to-binder ratio.

For the base mortar, the binder mass was calculated using:

$$m_b \left(\frac{1}{SG_b} + \frac{w}{b} + \frac{s/b}{SG_s} \right) = 990$$

where m_b is the binder mass, SG_b is the binder specific gravity, SG_s is the sand specific gravity, w/b is the water-to-binder ratio, and s/b is the sand-to-binder ratio. After Phase 1 optimization, the water-to-binder ratio and sand-to-binder ratio were fixed as 0.425 and 2.25, respectively, for the modified mixes.

Rice Husk Ash was added as a mass replacement of cement. Therefore, the cement and RHA masses were calculated as:

$$m_{RHA} = r \times m_b$$

$$m_{cement} = (1 - r) \times m_b$$

where r is the RHA replacement fraction. The suitability of RHA as a pozzolanic material was checked using Loss on Ignition and reactive oxide content calculations according to pozzolan testing and specification requirements [18], [19]. In the thesis calculation, the RHA showed a Loss on Ignition of 0.83% and a combined reactive oxide content of 93.8%, confirming its pozzolanic suitability.

EPS was added as a volumetric replacement of M-sand because its density was much lower than that of normal fine aggregate. The EPS volume and remaining sand volume were calculated using:

$$V_{EPS} = v \times V_{s,total}$$

$$V_s = (1 - v) \times V_{s,total}$$

where v is the EPS volume replacement fraction. This method maintained the total aggregate volume constant while reducing the density of the mortar. For coated EPS, the increased bulk density after coating was considered in the volume-to-mass conversion.

For PP fiber mixes, fibers were treated as a volumetric addition to the optimized RHA-coated EPS mortar. Therefore, the fiber volume was included in the total volume balance:

$$V_b + V_w + V_s + V_{EPS} + V_{fiber} = 0.99$$

The required fiber mass was calculated as:

$$m_{fiber} = V_{fiber} \times SG_{PP} \times 1000$$

Fresh mortar flow was calculated using the flow table spread diameter according to the flow-table consistency method [20]. Flexural and compressive strengths were calculated using standard mortar prism strength equations [21]. The flexural strength was calculated as:

$$f_r = \frac{1.5F_f L}{b^3}$$

where F_f is the flexural failure load, L is the support span, and b is the prism width. Compressive strength was calculated using:

$$f_c = \frac{F_c}{1600}$$

where F_c is the maximum crushing load and 1600 mm² is the loading area of the broken prism half.

Water absorption and apparent porosity were calculated using oven-dry and saturated surface-dry masses following absorption and void calculation principles [22]. These calculations were used to compare the influence of RHA, coated EPS, and PP fiber on density, strength, water absorption, porosity, and toughness of the modified cement mortar.

IV. RESULTS AND DISCUSSION

The experimental results were discussed phase by phase to show the contribution of each modification: base matrix optimization, density reduction using coated EPS, RHA-based pozzolanic densification, EPS coating validation, and PP fiber crack-bridging. Strength values were interpreted using the standard mortar prism strength procedure, while water absorption and porosity were used to judge the internal quality of the hardened matrix.

A. Phase-wise Summary of Key Outcomes

Table 1 : Condensed phase-wise results recreated from the thesis experimental data

Phase	Main variable	Key result	Interpretation
1	w/b and s/b	Selected w/b = 0.425, s/b = 2.25; f_c = 35.58 MPa; f_r = 4.75 MPa	Practical balance of strength, workability, density and porosity.
2	RHA + coated EPS	Optimum: 10% RHA + 20% coated EPS; density approx. 1993 kg/m ³ ; f_c approx. 16.78 MPa	Moderate EPS reduced weight while RHA reduced porosity and improved matrix quality.
3	RHA chemical effect	Ca/Si reduced from 24.85 to 5.65 at ITZ	RHA improved ITZ chemistry by reducing calcium-rich weak phases.
4	EPS coating	At 20% EPS, f_c improved 36.45% and f_r improved 64.29%	Coating improved EPS-paste interlock and reduced interfacial weakness.
5	PP fiber dosage	0.5% PP gave the best balance: f_c = 15.62 MPa, f_r = 3.59 MPa	Low fiber content improved crack resistance without excessive void formation.

B. Base Mortar Optimization (Phase 1)

Phase 1 established the reference mortar before adding RHA, EPS or PP fiber. The lowest water-to-binder ratio produced the highest strength, but it also made the mix too stiff for later

additions. Therefore, the center-point mix was selected instead of the maximum-strength mix. This mix used w/b = 0.425 and s/b = 2.25, giving an average compressive strength of 35.58 MPa, flexural strength of 4.75 MPa, 28-day density of about 2271 kg/m³ and apparent porosity of about 7.2%. This result agrees with absolute-volume mix proportioning logic, where strength and practical workability must be balanced [13].

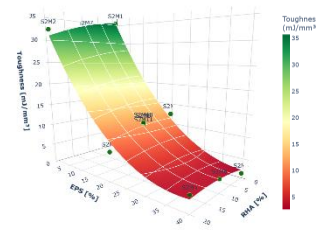


Figure 8 : Response Surface/Contour Plots for Toughness – Phase 2

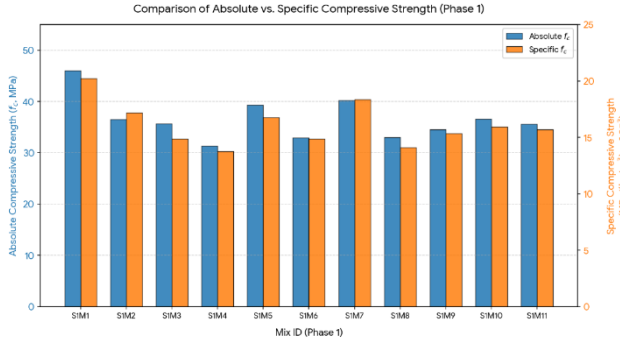


Figure 5 : Phase 1 absolute and specific compressive strength comparison

C. Lightweight Aggregate and RHA Optimization (Phase 2)

Phase 2 introduced coated EPS as the lightweight aggregate and RHA as the pozzolanic binder replacement. Increasing EPS clearly reduced density, but high EPS content caused a major reduction in compressive strength and toughness. The 40% EPS mixes reached densities near 1719-1748 kg/m³, but their compressive strength dropped to about 4-5 MPa. This confirms that EPS acts as a low-stiffness inclusion and weakens the continuous cementitious skeleton when used at high volume.

The best compromise was obtained using 10% RHA and 20% coated EPS. This condition reduced density to about 1993 kg/m³ while retaining about 16.78 MPa compressive strength and 3.26 MPa flexural strength. The apparent porosity remained close to the dense control, showing that the RHA helped compensate for the porosity introduced by EPS. This behaviour is consistent with RHA literature, where reactive silica improves strength and permeability by secondary C-S-H formation.

Table 2 : Key Phase 2 data showing density reduction and strength loss with increasing EPS

Condition	RHA (%)	EPS (%)	f _c (MPa)	rho28d (kg/m ³)	Porosity (%)	Toughness
Dense control	0	0	35.58	2287	6.04	35.477
RHA only	10	0	48.93	2263	5.83	32.945
EPS only	0	20	12.25	2011	7.65	12.578
Optimum balance	10	20	16.78	1993	6.12	11.58
High EPS	10	40	5.21	1734	6.94	3.844

D. Microstructural and Chemical Interfacial Transition Zone (ITZ) Diagnostics (Phase 3)

Phase 3 checked whether the strength improvement from RHA was supported by a chemical change at the interfacial transition zone (ITZ). The comparison used 40% EPS mixes with and without 20% RHA. The average Ca/Si ratio reduced from 24.85 to 5.65 after RHA addition, which indicates that the EPS boundary became less calcium-rich and more silica-rich. This supports the pozzolanic interpretation that RHA consumed calcium hydroxide and supported secondary C-S-H formation

Table 3 : Comparative EDS Elemental Analysis and ITZ Ca/Si Ratios (3-Location Average)

Mix ID	EPS (%)	RHA (%)	Avg. Ca (at. %)	Avg. Si (at. %)	Avg. Ca/Si Ratio	ITZ State Characteristic
S3M1	40	0	38.27	1.54	24.85	Weak / Highly Crystalline
S3M2	40	20	21.85	3.87	5.65	Dense / Amorphous C-S-H

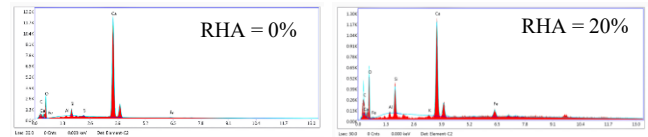


Figure 9 : EDS Spectra Mapping comparing elemental peaks of S3M1 and S3M2 at the ITZ

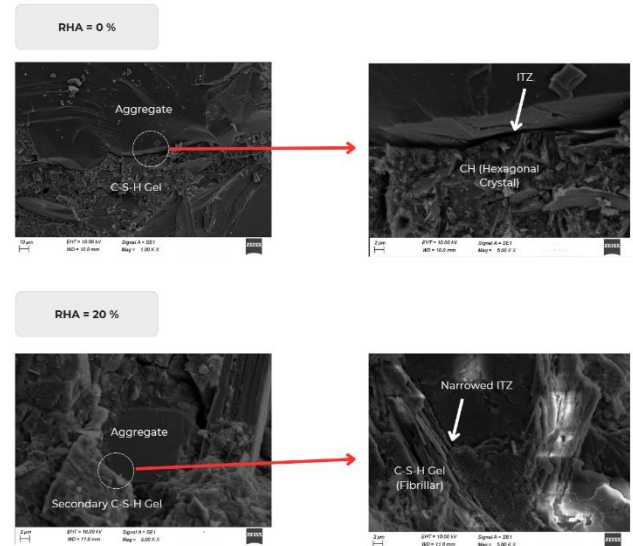


Figure 10 : SEM Micrographs demonstrating the ITZ morphology of S3M1 (top) showing boundary gaps/crystals, and S3M2 (bottom) showing the dense C-S-H gel formation

E. Physical Coating Effect of EPS (Phase 4)

Phase 4 isolated the influence of the PVAc/M-sand coating. At 20% EPS, coating increased compressive strength from 10.15 MPa to 13.85 MPa and flexural strength from 2.10 MPa to 3.45 MPa. At 40% EPS, the coating effect was even stronger, increasing compressive strength from 4.20 MPa to 7.15 MPa. These results show that the coating reduced EPS-paste debonding by roughening the particle surface and

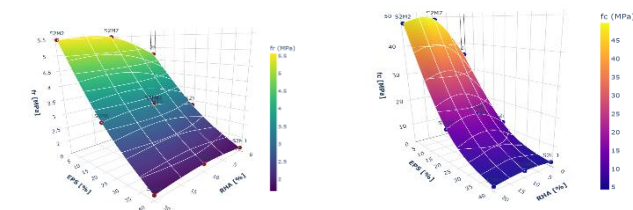


Figure 6 : Response Surface/Contour Plots for Compressive Strength – Phase 2

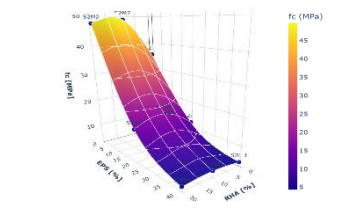


Figure 7 : Response Surface/Contour Plots for Toughness – Phase 2

improving mechanical interlocking, which is necessary because raw EPS behaves like a weak internal void.

Table 4 : Microstructural and coating-validation results from Phases 3 and 4

Phase	Comparison	Main change	Result meaning
3	0% RHA vs 20% RHA at 40% EPS	Ca/Si: 24.85 -> 5.65	Less CH-rich ITZ and more silica-rich hydrated phase
4	20% EPS uncoated vs coated	fc: 10.15 -> 13.85 MPa; fr: 2.10 -> 3.45 MPa	Coating improved mechanical interlock
4	40% EPS uncoated vs coated	fc: 4.20 -> 7.15 MPa; fr: 0.95 -> 1.85 MPa	Coating became more important at high EPS volume

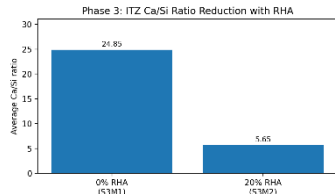


Figure 7 : Phase 3 Ca/Si reduction at the ITZ. Source: chart generated from thesis EDS results.

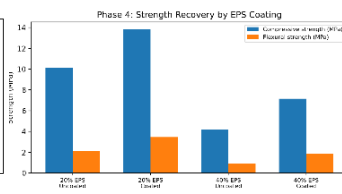


Figure 6 : Phase 4 strength recovery due to EPS coating. Source: chart generated from thesis results

F. PP Fiber Toughness Optimization (Phase 5)

Phase 5 introduced PP micro-fibers into the optimized coated-EPS matrix. The fiber mainly improved flexural response by bridging cracks and delaying brittle fracture. At 0.5% fiber, flexural strength rose from 3.26 MPa to 3.59 MPa and toughness increased from 14.71 to 18.24 mJ/mm³, while compressive strength eased slightly from 16.78 MPa to 15.62 MPa. Flexural strength continued rising to a 4.00 MPa peak at 1.0% fiber, then fell back to 3.65 MPa at 1.5% and 2.89 MPa at 2.0%, as higher dosages caused fiber balling that trapped voids and drove compressive strength down further, to 11.85 MPa and 11.06 MPa respectively. Therefore, 0.5% PP fiber was selected as the practical optimum, balancing toughness gains against strength retention. This trend agrees with fiber-reinforced cementitious composite behavior reported in the literature.

Table 5 : Phase 5 fiber optimization data from the thesis results

PP fiber (%)	fc (MPa)	fr (MPa)	Observation
0.0	16.78	3.26	Brittle baseline
0.5	15.62	3.59	Best overall balance
1.0	14.14	4.00	Flexural peak, compression declining
1.5	11.85	3.65	Flexural falling back, compression still declining
2.0	11.06	2.89	Fiber balling dominates, both properties reduced

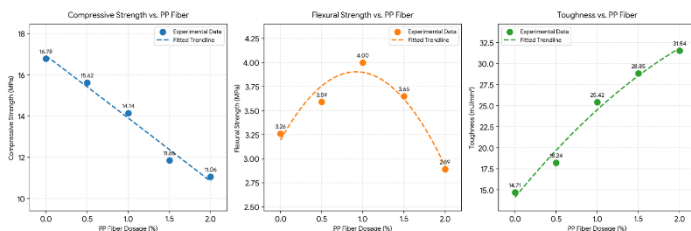


Figure 13 : Effect of PP fiber dosage on compressive flexural strength and toughness. Source: chart generated from thesis Phase 5 results.

G. Overall Discussion

The results confirm that the modified mortar performance was controlled by a balance between three mechanisms. Coated EPS reduced density, but excessive EPS weakened the continuous load-carrying skeleton. RHA improved the paste and ITZ by pozzolanic reaction, but it could not fully recover the strength loss caused by very high EPS volume. PP fiber improved flexural resistance and crack control, but only up to the dosage where dispersion remained good. Therefore, the most suitable mix was not the one with maximum EPS, maximum RHA or maximum fiber, but the balanced ternary combination.

Based on the complete phase-wise evaluation, the recommended final combination is $w/b = 0.425$, $s/b = 2.25$, 10% RHA, 20% PVAc/M-sand coated EPS and 0.5% PP fiber. This mix reduced density compared with the plain mortar, retained useful strength for lightweight precast applications, improved ITZ quality through RHA, strengthened EPS bonding through coating and improved post-cracking behavior through PP fiber bridging.

V. CONCLUSION AND FUTURE WORK

This study successfully developed a lightweight modified EPS cement mortar by combining rice husk ash, surface-coated expanded polystyrene beads, and polypropylene micro-fibers. The findings confirm that these three materials work as a complementary system rather than separate additives. RHA improved the cement matrix through pozzolanic reaction and pore refinement, coated EPS reduced density while improving aggregate–paste bonding, and PP fibers enhanced crack-bridging and post-cracking toughness. The optimized ternary mix containing 10% RHA, 20% coated EPS, and 0.5% PP fiber achieved 15.62 MPa compressive strength, 3.59 MPa flexural strength, 18.24 mJ/mm³ toughness, 5.79% apparent porosity, and 1993 kg/m³ density, while reducing self-weight by nearly 300 kg/m³ compared with conventional dense mortar. Therefore, the developed mortar can be considered suitable for sustainable non-load-bearing precast partitioning applications.

Future work should focus on validating the long-term durability and practical performance of the optimized mortar under real service conditions. Tests such as chloride penetration, sorptivity, sulfate resistance, drying shrinkage, and weathering resistance should be carried out. Further studies should also evaluate thermal conductivity and acoustic performance, since EPS can improve insulation properties. Finally, the optimized mix should be scaled up from laboratory prisms to full-size precast wall panels and tested under handling, transportation, impact, and installation loads to confirm its suitability for industrial production.

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